



Algebra liniova 2026

$$x^2 = 4$$

$$\checkmark \rightsquigarrow x = 2, -2 \quad \sqrt{4}$$

$$\text{liniova} \left\{ \begin{array}{l} 3x + 8 = 0 \rightsquigarrow x = -\frac{8}{3} \\ 20 - 13x = 2 \rightsquigarrow x = \frac{18}{13} \end{array} \right.$$

$$\sin^2 d + \cos^2 d = 1 \rightsquigarrow \text{wspyt, t\u00e9to l\u00edn\u00ed\u00ed nevy\u00e1\u00e1\u00e1}$$

$$\begin{array}{ccccc} x^2 & + & 6x & + & 4 = 0 \rightsquigarrow x = -3 \pm \sqrt{5} \quad \checkmark \\ a & & b & & c \end{array}$$

$$\nearrow \underline{\underline{x^2 = 5}}$$

$$2^x = 3 \quad x = \log_2 3$$

$$\left\{ \begin{array}{l} 2x + 4y = 10 \rightsquigarrow 2x = 10 - 4y \rightsquigarrow \underline{\underline{x = 5 - 2y}} \\ -3\underline{\underline{x}} + 2y = 9 \end{array} \right.$$

$$-3(5 - 2y) + 2y = 9$$

$$-15 + 6y + 2y = 9$$

$$8y = 24$$

$$y = 3$$

$$x = 5 - 2 \cdot 3 = -1$$

$$2x_1 + 4x_2 = 10$$

$$\cdot 3 \quad 2 \cdot (-1) + 4 \cdot 3 = 10 \quad \cdot 3$$

$$6 \cdot (-1) + 12 \cdot 3 = 30$$

Jesli lezy i prawy stronq pomnozimy przez te same liczby, to uklad te same x_1, x_2 dostajemy

$$2 \cdot x_1 + 4x_2 = 10$$

$$\rightarrow -3x_1 + 2x_2 = 9$$

uklad powoda

$$2x_1 + 4x_2 + (-3x_1 + 2x_2) = 19$$

$$\rightarrow -x_1 + 6x_2 = 19$$

Jesli lewe stronq dodamy do siebie i prawe stronq dodamy do siebie, to te same x_1, x_2 dostajemy.

$$\underline{2x_1 + 4x_2 = 10} \quad \cdot \frac{1}{2} \quad \leadsto \underline{x_1 + 2x_2 = 5} \leftarrow$$

$$\underline{-3x_1 + 2x_2 = 9} \quad \underline{-3x_1 + 2x_2 = 9} \leftarrow$$

$3R_1 + R_2$

$$x_1 + 2x_2 = 5$$

$$\cancel{-3x_1 + 2x_2 = 9}$$

$$0x_1 + 8x_2 = 24 \quad \cdot \frac{1}{8} \quad \leadsto x_2 = 3$$

$$x_1 + 2 \cdot 3 = 5 \leadsto x_1 = -1$$

$$\begin{array}{cc} x_1 & x_2 \\ \left[\begin{array}{cc|c} 2 & 4 & 10 \\ -3 & 2 & 9 \end{array} \right] & \xrightarrow{R_1 \rightarrow \frac{1}{2}R_1} & \left[\begin{array}{cc|c} 1 & 2 & 5 \\ -3 & 2 & 9 \end{array} \right] & \xrightarrow{R_2 \rightarrow R_2 + 3R_1} \end{array}$$

$$\rightarrow \left[\begin{array}{cc|c} 1 & 2 & 5 \\ 0 & 8 & 24 \end{array} \right] \xrightarrow{R_2 \rightarrow \frac{1}{8}R_2} \left[\begin{array}{cc|c} 1 & 2 & 5 \\ 0 & 1 & 3 \end{array} \right]$$

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$$x_2 = 3$$

$$x_1 + 2 \cdot 3 = 5$$

$$x_1 = -1$$

$$x_1 + 2x_2 + 3x_3 = 11$$

$$2x_1 + x_2 + x_3 = 7$$

$$-x_1 + 4x_2 + 2x_3 = 4$$

$$\begin{array}{l} \text{0} \rightarrow \\ \text{0} \rightarrow \end{array} \left[\begin{array}{ccc|c} 1 & 2 & 3 & 11 \\ 2 & 1 & 1 & 7 \\ -1 & 4 & 2 & 4 \end{array} \right] \xrightarrow{\begin{array}{l} R_2 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow R_3 + R_1 \end{array}} \left[\begin{array}{ccc|c} 1 & 2 & 3 & 11 \\ 0 & -3 & -5 & -15 \\ 0 & 6 & 5 & 15 \end{array} \right]$$

$$\xrightarrow{R_2 \rightarrow -\frac{1}{3}R_2} \left[\begin{array}{ccc|c} 1 & 2 & 3 & 11 \\ 0 & 1 & \frac{5}{3} & 5 \\ 0 & 6 & 5 & 15 \end{array} \right] \xrightarrow{R_3 \rightarrow R_3 - 6R_2} \left[\begin{array}{ccc|c} 1 & 2 & 3 & 11 \\ 0 & 1 & \frac{5}{3} & 5 \\ 0 & 0 & -5 & -15 \end{array} \right]$$

$$R_3 \rightarrow -\frac{1}{5} R_3 \quad \begin{bmatrix} 1 & 2 & 3 & | & 11 \\ 0 & 1 & \frac{5}{3} & | & 5 \\ 0 & 0 & 1 & | & 3 \end{bmatrix} \begin{array}{l} \rightarrow x_1 + 2 \cdot 0 + 3 \cdot 3 = 11 \\ \quad \quad \quad \rightarrow x_1 = 2 \\ \rightarrow x_2 + \frac{5}{3} \cdot 3 = 5 \\ \quad \quad \quad \rightarrow x_2 = 0 \end{array}$$

$$\rightarrow x_3 = 3$$

$$\begin{aligned} x_1 + x_2 - x_3 &= 1 \\ 2x_1 - 2x_2 + 10x_3 &= 2 \end{aligned}$$

$$3x_1 + x_2 + 4x_3 = 4$$

$$\begin{array}{c} \rightarrow \\ \rightarrow \\ 0 \\ 0 \end{array} \begin{bmatrix} 1 & 1 & -1 & | & 1 \\ 2 & -2 & 10 & | & 2 \\ 3 & 1 & 4 & | & 4 \end{bmatrix} \begin{array}{l} R_2 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow R_3 - 3R_1 \end{array} \begin{bmatrix} 1 & 1 & -1 & | & 1 \\ 0 & -4 & 12 & | & 0 \\ 0 & -2 & 7 & | & 1 \end{bmatrix}$$

$$R_2 \rightarrow -\frac{1}{4} R_2 \quad \begin{bmatrix} 1 & 1 & -1 & | & 1 \\ 0 & 1 & -3 & | & 0 \\ 0 & -2 & 7 & | & 1 \end{bmatrix} \xrightarrow{R_3 \rightarrow R_3 + 2R_2}$$

$$\begin{bmatrix} 1 & 1 & -1 & | & 1 \\ 0 & 1 & -3 & | & 0 \\ 0 & 0 & 1 & | & 1 \end{bmatrix} \quad \begin{array}{l} x_1 = -1 \\ x_2 = 3 \\ x_3 = 1 \end{array}$$

$$\begin{cases} x_1 + x_2 + 3x_3 + x_4 = 4 \\ 2x_1 - x_2 - x_3 + 2x_4 = 5 \\ x_1 + 2x_2 + 2x_3 - x_4 = 1 \\ -x_1 + x_2 + x_3 + 3x_4 = 6 \end{cases}$$